

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

IT-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

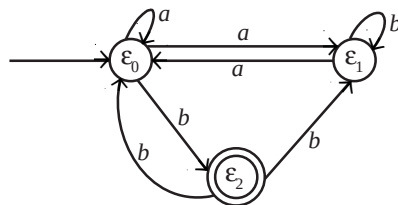
Theory of Computation

Time : Three Hours

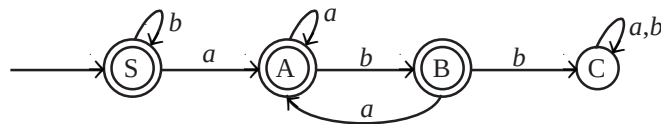
Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Design DFA which accepts the language
 $h = \{ 0, 000, 00000, 0000000, \dots \}$ over the alphabet $\{0\}$.
b) Design NFA to accept strings with a 's and b 's such that the string end with ' aa '.
2. a) Construct a DFA equivalent to NFA



- b) Distinguish between Moore and Melay Machines.
3. Construct a regular grammar for the following FA.



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[2]

4. a) What is Regular expression and regular set? Explain with an example.
b) Obtain the grammar to generate the language
$$h = \{0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0\}$$
5. Consider the grammar $G = (\{A_1, A_2, A_3\}, \{a, b\}, \{p\}, \{A_1\})$ where p consist of following production rules.
$$A_1 \rightarrow A_2 A_3$$
$$A_2 \rightarrow A_3 A_1 / b$$
$$A_3 \rightarrow A_1 A_2 / a$$

Convert in to GNF.
6. Construct a PDA for the following language
i) $L = \{a^n b^m c^m d^n : m, n \geq 1\}$
ii) $L = wcw^R \quad w \in (a, b)^2$
7. a) Design a Turing Machine for multiplication of two integers.
b) Explain Recursive and Recurively enumerable language.
8. Write short notes (any three)
a) NP type problem
b) Decidable problem
c) Pumping lemma
d) CFG
